

Technology Stack Details

Date:	3 October 2026	Team ID:	unm1611MADHANRAJ
Project Name:	AI FitTrack API	Maximum Marks:	2 Marks

Architecture Component / Layer Details — Part 1:

S.No	Component / Layer	Technology Chosen	Justification / Purpose
1	Runtime Environment	Node.js (v20+ LTS)	High-performance event-driven JavaScript runtime supporting non-blocking asynchronous operations for concurrent user traffic.
2	Web Framework	Express.js (v4.x)	Minimalist, robust web application framework offering flexible middleware routing, HTTP body parsing, and route modularity.
3	Database Engine	MongoDB Atlas (Cloud)	Fully managed cloud document database with dynamic schema agility, horizontal scaling, and flexible JSON document storage.

Architectural Stack Rationale:

The selection of Node.js, Express, MongoDB Atlas, and Google Gemini AI provides a unified JavaScript/JSON ecosystem. This ensures rapid development, seamless data flow between layers, and state-of-the-art Generative AI capabilities without heavy infrastructure overhead.

Technology Stack Details — Part 2

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Architecture Component / Layer Details — Part 2:

S.No	Component / Layer	Technology Chosen	Justification / Purpose
4	Object Data Modeling	Mongoose ODM (v8.x)	Provides structured schema definitions, data type validation, default timestamps, and relational ObjectId referencing.
5	Generative AI Engine	Google Gemini 1.5 Flash	Cutting-edge lightweight multimodal LLM delivering high reasoning speeds and accurate workout routine formulations.
6	Gen AI Integration	@google/genai SDK (v2.x)	Official Google client library providing clean programmatic interfaces for prompt dispatch and model response handling.
7	Security & Tokenization	jsonwebtoken (JWT)	Enables stateless, secure user authentication with signed HMAC-SHA256 tokens passed via Authorization Bearer headers.
8	Password Hashing	bcryptjs (v2.4.3)	Cryptographic hashing function utilizing 10 salt rounds to defend user passwords against rainbow tables and brute force attacks.

Technology Stack Details — Part 3 & Dependency Matrix

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Architecture Component / Layer Details — Part 3:

S.No	Component / Layer	Technology Chosen	Justification / Purpose
9	HTTP Logging	morgan (v1.10.0)	HTTP request logging middleware capturing method, URL, status code, and response latency in the server console.
10	Cross-Origin Policy	cors (v2.8.5)	Enables secure cross-origin resource sharing allowing web and mobile frontend applications to consume the REST API.
11	Environment Config	dotenv (v16.x)	Loads sensitive environmental secrets (PORT, MONGO_URI, JWT_SECRET, GEMINI_API_KEY) cleanly into process.env.
12	API Testing Suite	Postman Collection v2.1	Automated API verification suite featuring environment variables, token chaining, and automated test assertion scripts.

Package Dependency Matrix:

Package Name	Version	Type	Core Responsibility in AI FitTrack API
express	^4.19.2	Production	HTTP server initialization, routing, and middleware mounting.
mongoose	^8.4.1	Production	MongoDB schema modeling and database connection handler.
@google/genai	^2.0.0	Production	Google Gemini 1.5 Flash SDK communication and prompt pipeline.
jsonwebtoken	^9.0.2	Production	HMAC-SHA256 JWT access token signing and verification.
bcryptjs	^2.4.3	Production	Salted password hashing and authentication comparison.
dotenv	^16.4.5	Production	Environment variable loader from .env file.
morgan	^1.10.0	Production	HTTP request latency and endpoint status logger.